

Enhanced sampling techniques

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

Day 3/5

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Ensemble averages

- **Ensemble averages:** given microscopic observable $O(\Gamma)$, its expected value is

$$\langle O \rangle = \int d\Gamma O(\Gamma) \mathcal{P}(\Gamma)$$

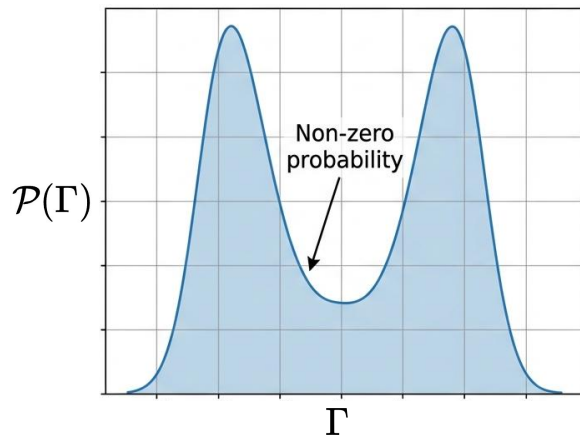
- **Ergodicity:** if $\Gamma(t)$ denotes a **single trajectory** starting from $\Gamma(0)$,

$$\lim_{T \rightarrow +\infty} \frac{1}{T} \int_0^T dt O(\Gamma(t)) = \langle O \rangle$$

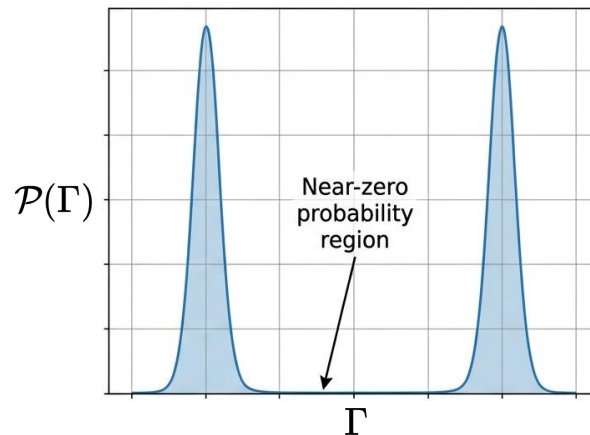
If the system is ergodic, ensemble averages (across different realizations) are equivalent to time averages (on a single realization)!

Ensemble averages

- In practice, when can we reliably estimate averages from a **finite trajectory**?
→ when the samples $\{\Gamma_i\}$ ($i = 1, \dots, N$) faithfully represent the distribution $\mathcal{P}(\Gamma)$
- This typically occurs when the distribution $\mathcal{P}(\Gamma)$ is “simple” enough



(a) Connected Modes
(Non-zero Intermediate Probability)



(b) Well-Separated Modes
(Near-zero Intermediate Probability)

Ensemble averages

- Let's suppose that $\mathcal{P}(\Gamma)$ is “simple enough” (or that the trajectory is “long enough” to explore all modes)
- $\langle O \rangle$ can be simply estimated from an average along the trajectory:

$$\langle O \rangle \simeq \frac{1}{N} \sum_{i=1}^N O_i$$

Diagram annotations:

- Arrow from N to "Total number of steps in the trajectory"
- Arrow from O_i to "Value of the microscopic observable at the i-th step"
- Arrow from i to "Step index"

- What is the **error** associated to $\langle O \rangle$?

applying the **central limit theorem**:

$$\text{Var}[\langle O \rangle] = \frac{\text{Var}[O_i]}{\sqrt{N}}$$

Diagram annotations:

- Arrow from $\text{Var}[O_i]$ to "(spread of $\{O_i\})$ "

What's wrong with this relationship?

Errors of time averages

- N.B. the relationship $\text{Var}[\langle O \rangle] = \frac{\text{Var}[O]}{\sqrt{N}}$ holds if the samples $\{O_i\}$ are **independent!**

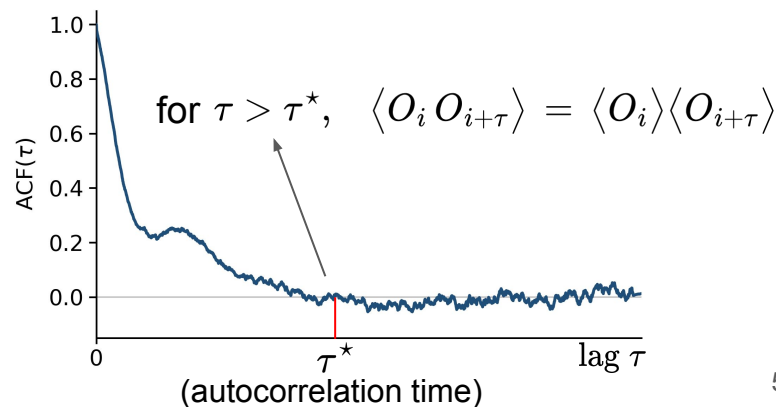
- By construction, points in a time series are **correlated through time**: for a generic lag τ ,

$$\langle O_i O_{i+\tau} \rangle \neq \langle O_i \rangle \langle O_{i+\tau} \rangle$$

- Degree of correlation though time quantified by the **autocorrelation function**:

$$\text{ACF}(\tau) = \frac{\langle O_i O_{i+\tau} \rangle - \langle O_i \rangle \langle O_{i+\tau} \rangle}{\text{Var}[O_i]}$$

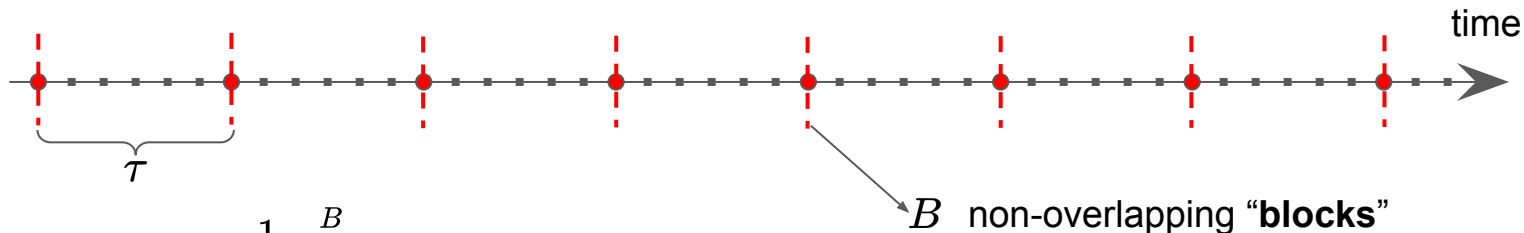
for a stationary dynamics (as in equilibrium MD)
it's only a function of τ



Errors of time averages

Two possibilities:

1. Subsample the time series with a stride equal (or larger) than the autocorrelation time



$$\langle O \rangle = \frac{1}{B} \sum_{i=1}^B O_i$$

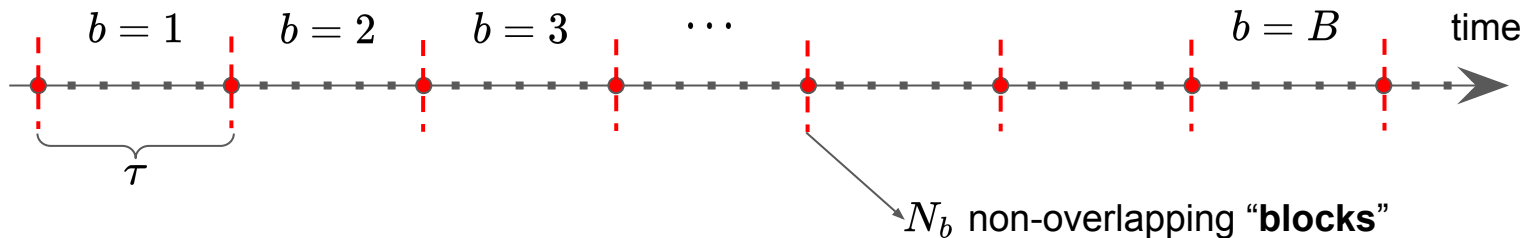
$$\text{Var}[\langle O \rangle] = \frac{\text{Var}[O_i]}{\sqrt{B}} \longrightarrow \text{Now this formula is correct, because samples are independent!}$$

Note that the autocorrelation time might be different for different observables!

Can we do better?

Errors of time averages

2. Block analysis



- Compute average over each block: $\langle O \rangle_b = \frac{1}{\tau} \sum_{i \in b} O_i$
- Treat these block-averages as our independent samples:

$$\langle O \rangle = \frac{1}{B} \sum_{b=1}^B \langle O \rangle_b$$

$$\text{Var}[\langle O \rangle] = \frac{\text{Var}[\langle O \rangle_b]}{\sqrt{N_b}}$$

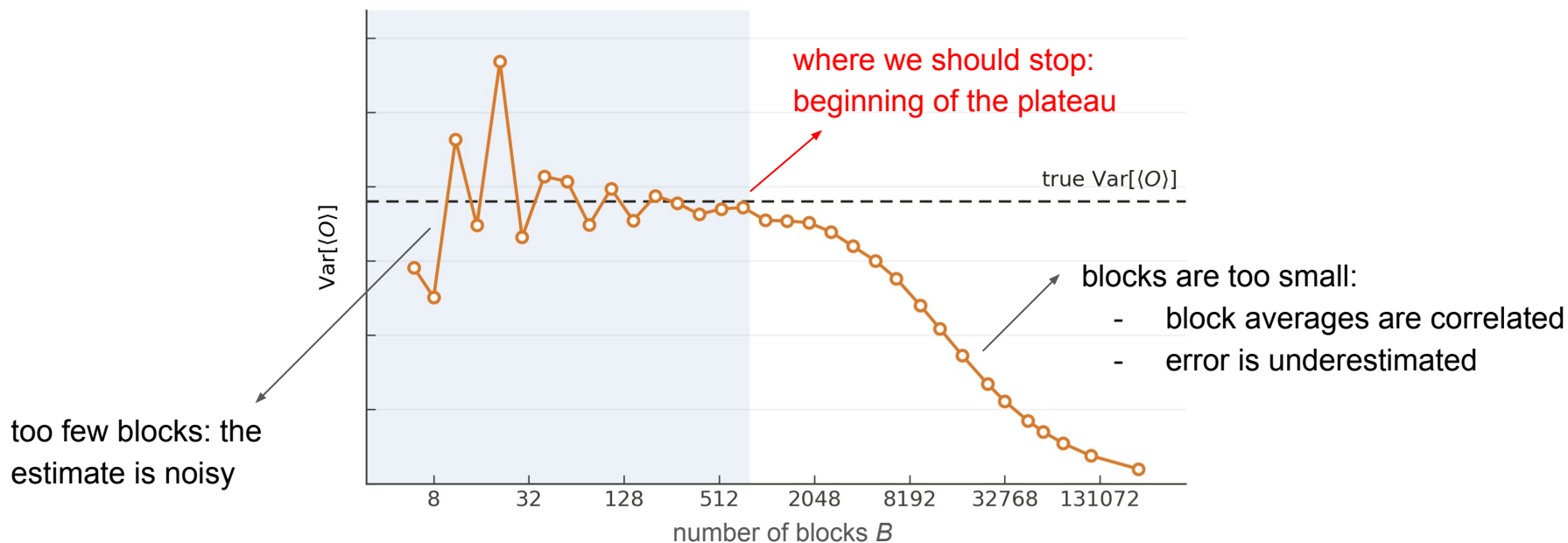
this is significantly smaller
than the variance of a single
sample! :)

- Problem: block-averages have longer time-correlation time than single samples...

Errors of time averages

2. Block analysis

- How can we identify correct block size to make the block averages independent?



Collective variables

A collective variable S is a low-dimensional (1D, 2D, ...) representation of the high-dimensional state Γ of the system:

$$S : \mathbb{R}^{6N} \rightarrow \mathbb{R}^d$$

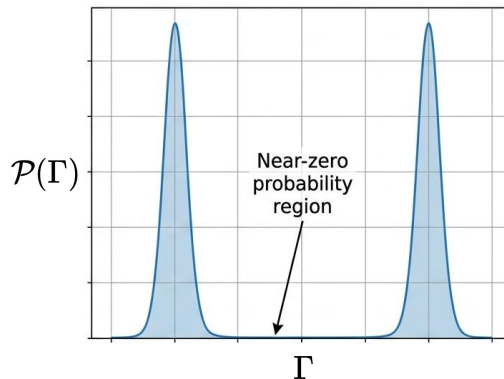
- Most CVs actually depend on the positions only: $S(\{\mathbf{q}_i\})$

Examples:

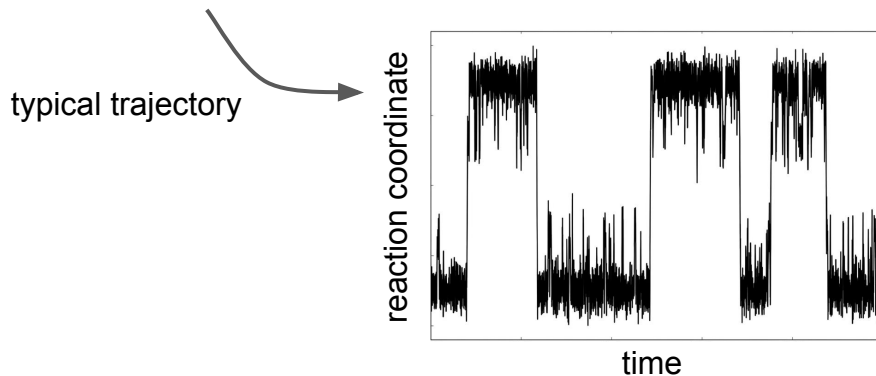
- Interatomic distance between two atoms
- Bond or dihedral angle
- Coordination number of an atom
- Radius of gyration
- Complex function of $\{\mathbf{q}_i\}$ learnt through dimensionality reduction
- ...

Collective variables

- Let's go back to the case of a multimodal probability distribution

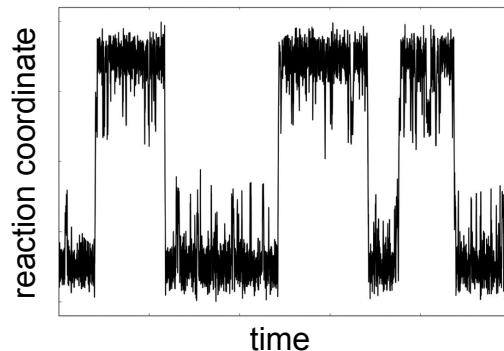


- A CV behaves as a **reaction coordinate** if it allows distinguishing “well” the two states (modes):



NB: not all CVs are good reaction coordinates!

Metastability and rare events



- Three relevant time scales:
 - “*relaxation*” time within a minimum (=metastable state) , $\tau_{\text{relaxation}}$
 - “*residence*” time between two transitions, $\tau_{\text{residence}}$
 - “*exchange*” time required to perform the transition, τ_{exchange}
 - **Metastability:** $\tau_{\text{relaxation}} \ll \tau_{\text{residence}}$
 - **Rare events:** $\tau_{\text{exchange}} \ll \tau_{\text{residence}}$
- } actually one implies the other, because $\tau_{\text{relaxation}} \sim \tau_{\text{exchange}}$

Free energies

- First: can we write the probability of finding the system in a given value of the CV $\mathcal{S}(\{\mathbf{q}_i\})$?

$$\mathcal{P}(\{\mathbf{q}_i\}) = \frac{1}{\mathcal{Z}} \exp\left(-\beta U(\{\mathbf{q}_i\})\right) \xrightarrow{?} \mathcal{P}(\mathcal{S})$$

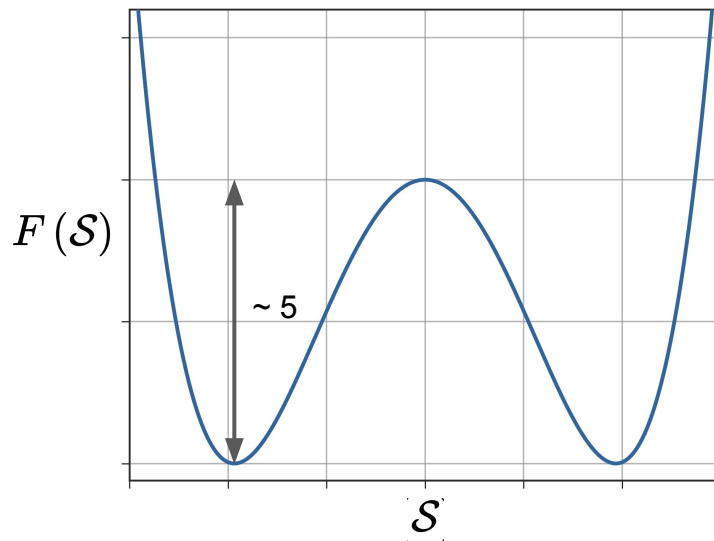
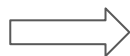
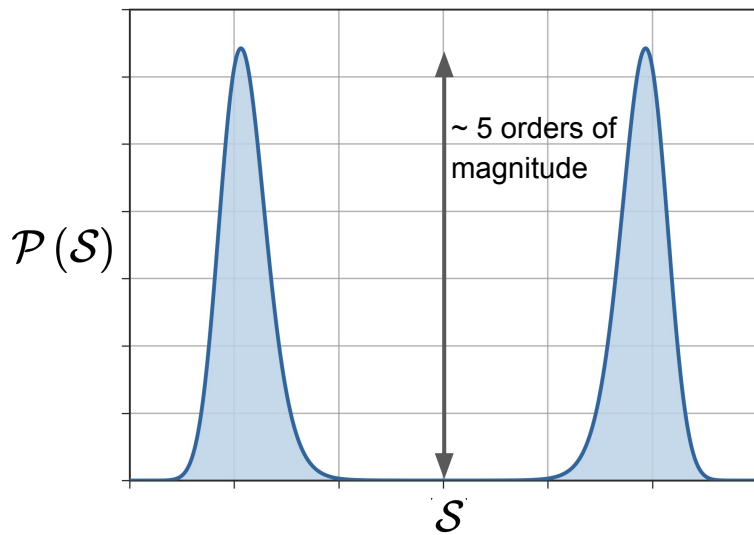
- Simpler notation from now on: $\mathbf{q} := \{\mathbf{q}_i\}$
- Using the delta-function:

$$\begin{aligned}\mathcal{P}(\mathcal{S} = s) &= \int d\mathbf{q} \, \delta(\mathcal{S}(\mathbf{q}) - s) \mathcal{P}(\mathbf{q}) \\ &= \frac{1}{\mathcal{Z}} \int d\mathbf{q} \, \delta(\mathcal{S}(\mathbf{q}) - s) \exp\left(-\beta U(\mathbf{q})\right)\end{aligned}$$

The **free energy** as a function of the CV $\mathcal{S}(\mathbf{q})$ is defined as $F(\mathcal{S}) = -\frac{1}{\beta} \log \mathcal{P}(\mathcal{S})$

Free energies

The **free energy** as a function of the CV $S(\mathbf{q})$ is defined as $F(S) = -\frac{1}{\beta} \log \mathcal{P}(S)$



- Modes of $\mathcal{P}(S)$ are **local minima** of $F(S)$
- Low-probability regions between modes become **barriers**

Sum up so far

- We want to sample a complex, high-dimensional target distribution $\mathcal{P}(\Gamma)$ (e.g. Boltzmann)
- In many cases, $\mathcal{P}(\Gamma)$ is multimodal and the modes are separated by low-probability regions: sampling it through a dynamics takes **long simulation times**

Why do we want to explore all the modes of $\mathcal{P}(\Gamma)$?

1. To compute ensemble averages through time averages
2. To study the transition mechanism itself! (e.g. how does a protein fold?)

- We suppose to now a “good” CV that behaves as a reaction coordinate (a variable that characterizes transitions between metastable states)



Assumption: the CV is able to capture in a high-dimensional space (e.g. 1D) the transition between two high-dimensional modes of $\mathcal{P}(\Gamma)$

Enhanced sampling

- Can we use our CV to speed up the exploration of $\mathcal{P}(\Gamma)$?

Enhanced sampling techniques aim at accelerating the “ergodic sampling” of $\mathcal{P}(\Gamma)$ by simulating rare transitions more frequently

- We will focus on two **CV-based enhanced sampling methods**:
 - Adaptive umbrella sampling
 - Metadynamics
- ...but the class of methods is much larger:
 - Steered MD
 - Replica-Exchange
 - Weighted-histogram analysis method (WHAM)
 - Transition path sampling
 -

Umbrella sampling

- **Idea:** simulate the system in a **biased potential** in which the barrier is lower (=more frequent transitions), then reweight biased distribution to get the **unbiased distribution**

Key assumption: add to the potential energy $U(\mathbf{q})$ a biased potential that depends on the state of the system through the CV $\mathcal{S}(\mathbf{q})$:

$$U(\mathbf{q}) \mapsto U(\mathbf{q}) + B(\mathcal{S}(\mathbf{q}))$$

- As a consequence the sampled distribution changes as well:

$$\mathcal{P}(\mathbf{q}) = \frac{1}{Z} \exp\left(-\beta U(\mathbf{q})\right) \longrightarrow \mathcal{P}_B(\mathbf{q}) = \frac{1}{Z_B} \exp\left(-\beta (U(\mathbf{q}) + B(\mathcal{S}(\mathbf{q})))\right)$$

Umbrella sampling

$$\mathcal{P}(\mathbf{q}) = \frac{1}{\mathcal{Z}} \exp\left(-\beta U(\mathbf{q})\right) \longrightarrow \mathcal{P}_B(\mathbf{q}) = \frac{1}{\mathcal{Z}_B} \exp\left(-\beta (U(\mathbf{q}) + B(\mathcal{S}(\mathbf{q})))\right)$$

- How are these two distributions related? Take the ratio:

$$\frac{\mathcal{P}(\mathbf{q})}{\mathcal{P}_B(\mathbf{q})} = \frac{\mathcal{Z}_B}{\mathcal{Z}} \exp\left(\beta B(\mathcal{S}(\mathbf{q}))\right)$$

$$\mathcal{P}(\mathbf{q}) = \frac{\mathcal{Z}_B}{\mathcal{Z}} \exp\left(\beta B(\mathcal{S}(\mathbf{q}))\right) \mathcal{P}_B(\mathbf{q})$$

- For the distribution of the CVs the same relationship holds:

$$\mathcal{P}(\mathcal{S}) = \frac{\mathcal{Z}_B}{\mathcal{Z}} \exp\left(\beta B(\mathcal{S})\right) \mathcal{P}_B(\mathcal{S})$$

Umbrella sampling

$$\mathcal{P}(\mathcal{S}) = \frac{\mathcal{Z}_{\mathcal{B}}}{\mathcal{Z}} \exp\left(\beta B(\mathcal{S})\right) \mathcal{P}_{\mathcal{B}}(\mathcal{S})$$

- What about the unbiased free energy? Substituting in $F(\mathcal{S}) = -\frac{1}{\beta} \log \mathcal{P}(\mathcal{S})$:

$$F(\mathcal{S}) = -\frac{1}{\beta} \log \frac{\mathcal{Z}_{\mathcal{B}}}{\mathcal{Z}} - B(\mathcal{S}) + F_{\mathcal{B}}(\mathcal{S})$$

↓
constant in $\mathcal{S}$

Umbrella sampling

- And **time averages**? Can we compute unbiased averages from the biased simulation?

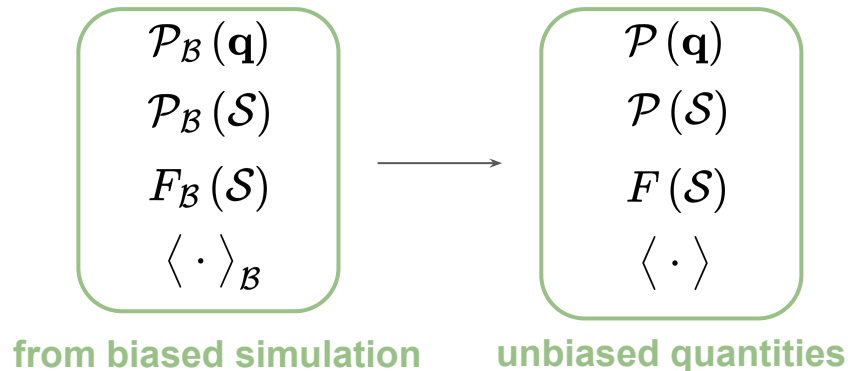
$$\begin{aligned}\langle O \rangle &= \int d\mathbf{q} O(\mathbf{q}) \mathcal{P}(\mathbf{q}) \\ &= \int d\mathbf{q} O(\mathbf{q}) \frac{Z_B}{Z} \underbrace{\exp\left(\beta B(\mathcal{S}(\mathbf{q}))\right)}_{w(\mathbf{q})} \mathcal{P}_B(\mathbf{q}) \\ &= \frac{Z_B}{Z} \int d\mathbf{q} O(\mathbf{q}) w(\mathbf{q}) \mathcal{P}_B(\mathbf{q}) = \frac{Z_B}{Z} \langle O w \rangle_B\end{aligned}$$

- Let's consider the constant function $O(\mathbf{q}) = 1$: $1 = \frac{Z_B}{Z} \langle w \rangle_B$

$$\langle O \rangle = \frac{\langle O w \rangle_B}{\langle w \rangle_B}$$

Umbrella sampling

- What we know now:



- What we **don't know**: how to choose the bias potential $B(\mathcal{S}(\mathbf{q}))$
- Desired feature: $B(\mathcal{S}(\mathbf{q}))$ should make the rare transitions less rare



In the language of free energy: it should **lower the free energy barrier** between the metastable states

Umbrella sampling

- Let's re-write the "Umbrella sampling identity", $F(\mathcal{S}) = -\frac{1}{\beta} \log \frac{Z_B}{Z} - B(\mathcal{S}) + F_B(\mathcal{S})$

$$F_B(\mathcal{S}) = F(\mathcal{S}) - B(\mathcal{S}) + c$$

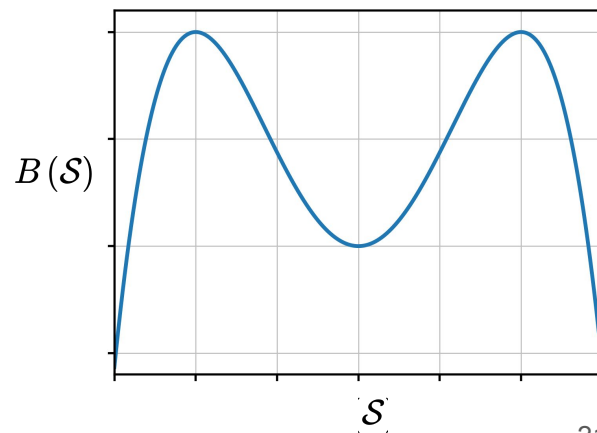
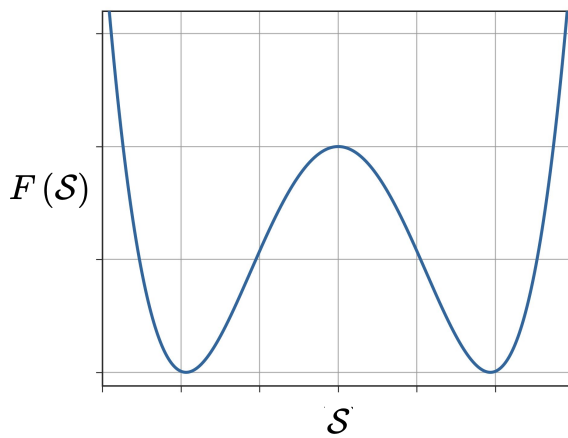
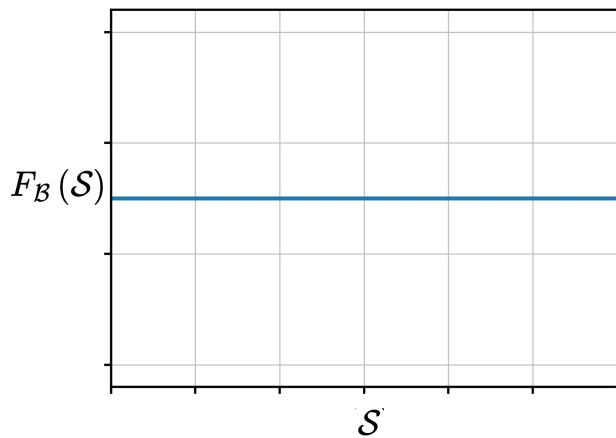
(Target) Biased free energy

=

Unbiased free energy

-

Bias potential



Umbrella sampling

The optimal bias is $B(\mathcal{S}) = -F(\mathcal{S})$

- Problem: we don't know $F(\mathcal{S})$, it's exactly what we are trying to sample!
- Different strategies to choose $B(\mathcal{S})$ bring to different CV-based enhanced sampling techniques; today we see two of them:
 - **Adaptive umbrella sampling***
 - **Metadynamics****
- Common goal: reach the condition $B(\mathcal{S}) = -F(\mathcal{S})$ iteratively by dynamically updating $B(\mathcal{S})$ from a starting guess

*M. Mezei, J. Comput. Phys. **68** (1987)

A. Laio and M. Parrinello, *Escaping free-energy minima*, PNAS **99 (2002)

Adaptive umbrella sampling

- **Goal:** reach the condition $B(\mathcal{S}) = -F(\mathcal{S})$ iteratively by dynamically updating $B(\mathcal{S})$ from a starting guess
- **Notation:** $\mathcal{B}^{(k)}(\mathcal{S})$, $F^{(k)}(\mathcal{S})$, $F_{\mathcal{B}}^{(k)}(\mathcal{S})$ denote quantities at step k
- **Algorithm:**

1) Initialize the bias potential: $\mathcal{B}^{(1)}(\mathcal{S}) = 0$

Run MD and compute the biased free energy: $F_{\mathcal{B}}^{(1)}(\mathcal{S})$

Compute best estimate of unbiased free energy: $F^{(1)}(\mathcal{S}) = -B^{(1)}(\mathcal{S}) + F_{\mathcal{B}}^{(1)}(\mathcal{S}) = F_{\mathcal{B}}^{(1)}(\mathcal{S})$

2) Update bias potential: $\mathcal{B}^{(2)}(\mathcal{S}) = -F^{(1)}(\mathcal{S})$

Run MD and compute the biased free energy: $F_{\mathcal{B}}^{(2)}(\mathcal{S})$ [...]

Compute best estimate of unbiased free energy: $F^{(2)}(\mathcal{S}) = -B^{(2)}(\mathcal{S}) + F_{\mathcal{B}}^{(2)}(\mathcal{S})$

Adaptive umbrella sampling

- **At iteration k:**

$$B^{(k)}(\mathcal{S}) = -F^{(k-1)}(\mathcal{S}) \quad \text{where} \quad F^{(k-1)}(\mathcal{S}) = -B^{(k-1)}(\mathcal{S}) + F_{\mathcal{B}}^{(k-1)}(\mathcal{S})$$

- Substituting we find the iterative update rule for the bias potential:

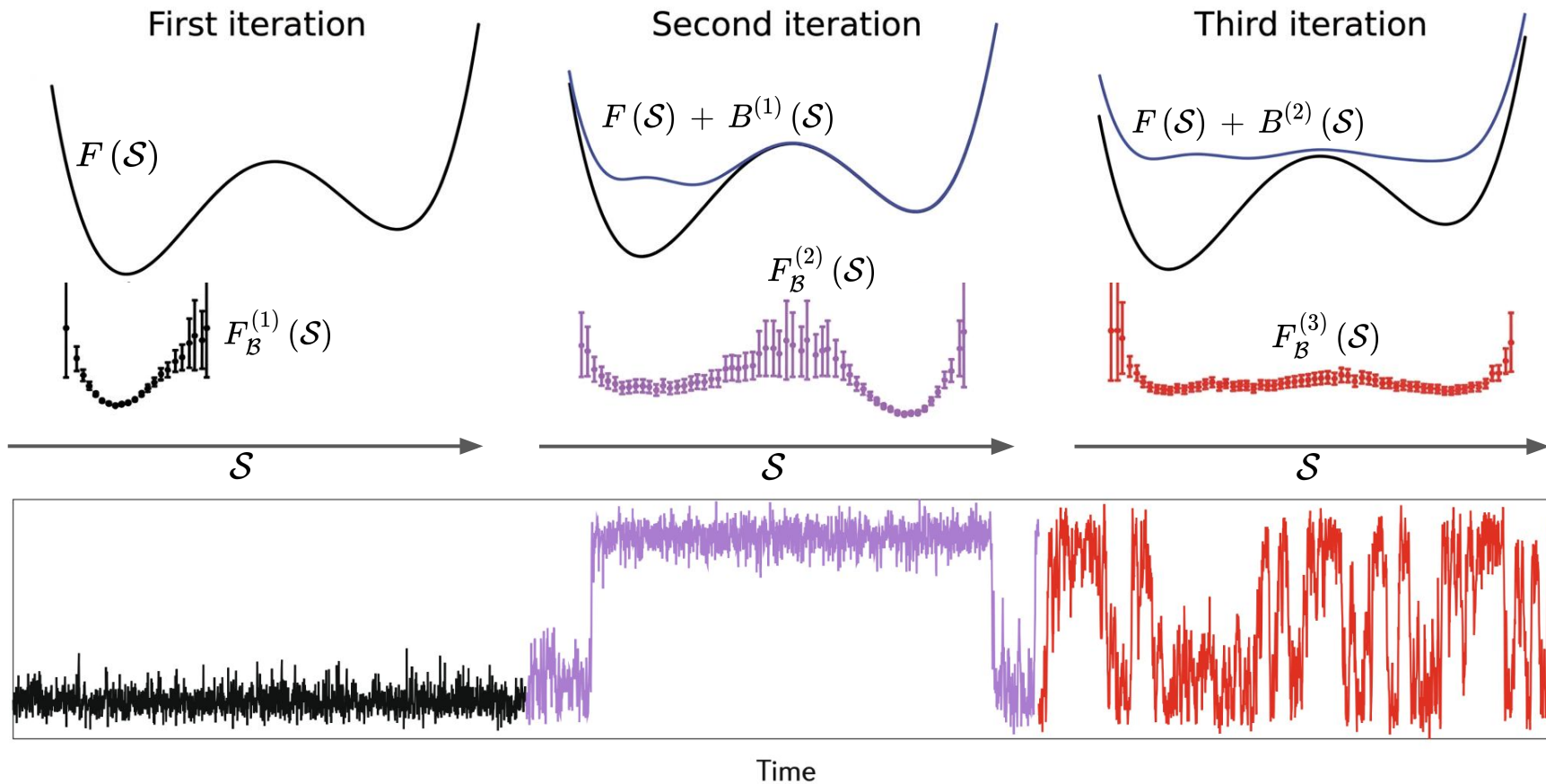
$$B^{(k)}(\mathcal{S}) = B^{(k-1)}(\mathcal{S}) - F_{\mathcal{B}}^{(k-1)}(\mathcal{S})$$

- **Stopping criterion:** when $B^{(k)}(\mathcal{S}) \simeq B^{(k-1)}(\mathcal{S}) (+const)$



(when $F_{\mathcal{B}}^{(k)}(\mathcal{S})$ is approximately flat)

Adaptive umbrella sampling



Adaptive umbrella sampling

Limitations

- Each iterations require a short MD run, but **long enough to “locally equilibrate”**

→ Equilibration time depends on the system!

- The biased free energy $F_B^{(k)}(\mathcal{S})$ at each step is computed with the following passages:

1. Compute a histogram over the CV: $\{\mathcal{P}_B^{(k)}(s_i)\}$
2. Compute the free energy on the grid points (bins): $\left\{\mathcal{F}_B^{(k)}(s_i) = -\frac{1}{\beta} \log \mathcal{P}_B^{(k)}(s_i)\right\}$
3. Compute a **smooth interpolation** of the free energy: $\mathcal{F}_B^{(k)}(s) = \mathcal{R}\left(\{\mathcal{F}_B^{(k)}(s_i)\}\right)$

→ not a trivial step, especially if the CV is multi-dimensional!

Metadynamics

- It can be seen as a “limit” of Adaptive Umbrella Sampling (not the way it was introduced)
- Adaptive Umbrella Sampling: at each iteration, run a short MD (length τ_{MD}) which is long enough to have a “decent” local estimate of the free energy
- Metadynamics: we don’t need a decent local estimate of the free energy, **we just want to fill the local minima and reach iteratively a flat biased free energy**:
 - we don’t need to reach any local equilibration: $\tau_{\text{MD}} \rightarrow 0$ (in practice extremely short simulations, ~ 500 steps)
 - we can update the bias based on a single-point “free energy”:

single value at iteration k

$$B^{(k)}(\mathcal{S}) = B^{(k-1)}(\mathcal{S}) - F_B^{(k-1)}(\mathcal{S})$$

Adaptive Umbrella Sampling

$$B^{(k)}(\mathcal{S}) = B^{(k-1)}(\mathcal{S}) + w \exp\left(-\frac{(\mathcal{S} - s^{(k)})^2}{2\sigma^2}\right)$$

Metadynamics

Metadynamics

$$B^{(k)}(\mathcal{S}) = B^{(k-1)}(\mathcal{S}) - F_B^{(k-1)}(\mathcal{S})$$

Adaptive Umbrella Sampling

$$B^{(k)}(\mathcal{S}) = B^{(k-1)}(\mathcal{S}) + w \exp\left(-\frac{(\mathcal{S} - s^{(k)})^2}{2\sigma^2}\right)$$

Metadynamics

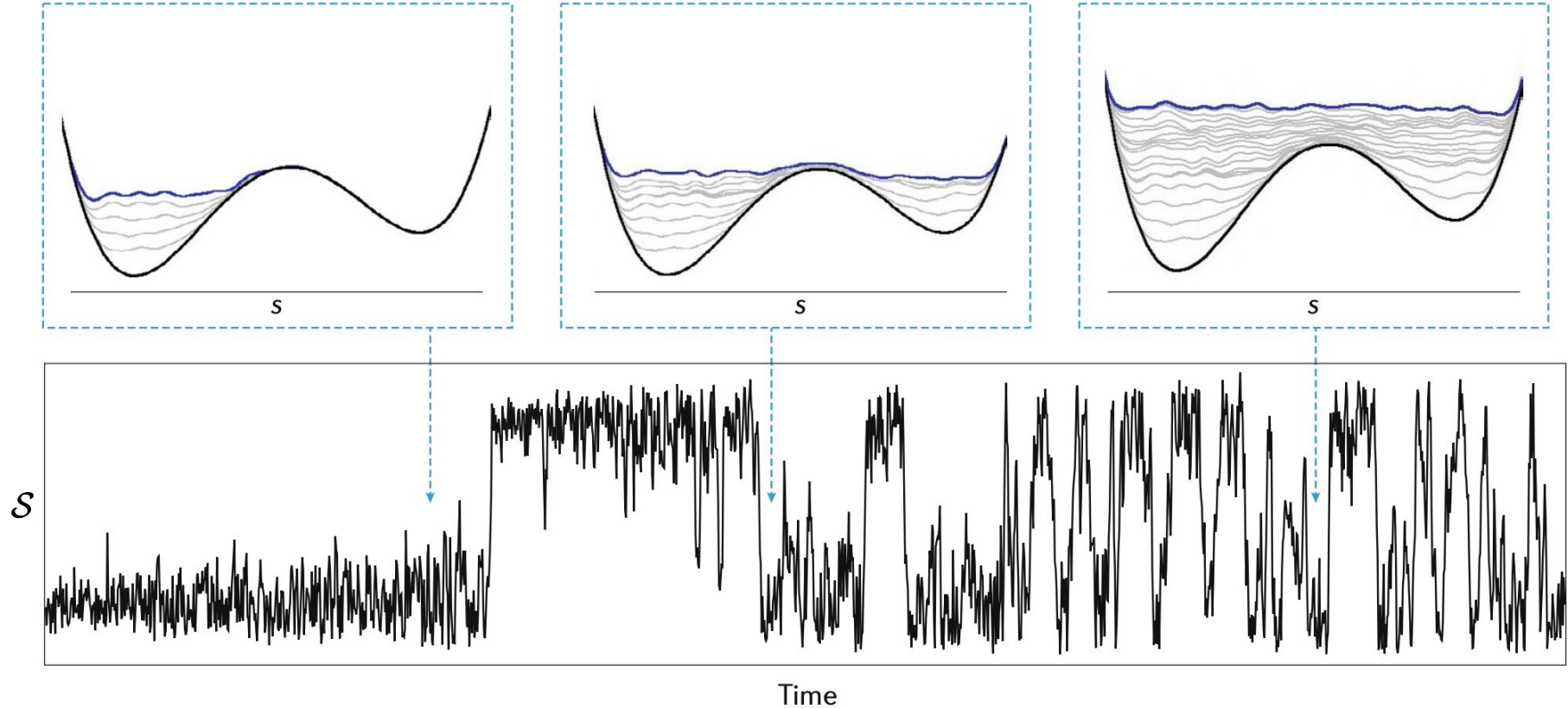
- It could be implemented even with a single step! However this would add a Gaussian which is too close to the previous one and slow down the “filling”
- w, σ^2 are **free parameters** which tune the impact of the update:
 - σ^2 sets the resolution of the final free energy (too large: low resolution; too small: slow filling)
 - w sets the impact of a single update: how much we perturb the system



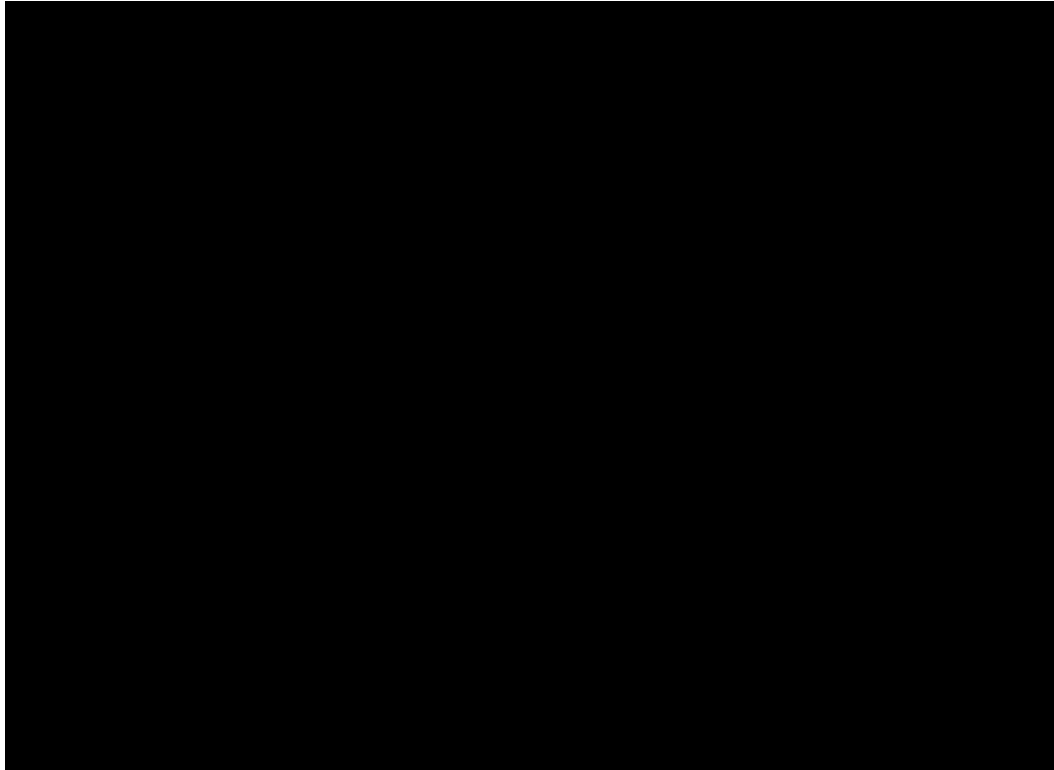
only important parameter is actually the ration w/τ_{MD}
(heuristic: $\tau_{\text{MD}} \simeq 1 \text{ ps}$; w : a fraction of $k_B T$)

can be set rigorously in an extension called “well-tempered metadynamics”

Metadynamics



Metadynamics



Metadynamics

Improvements wrt Adaptive Umbrella Sampling:

- No need to wait a local equilibration
- No need to perform any “histogram smoothing” to compute free energy at each step

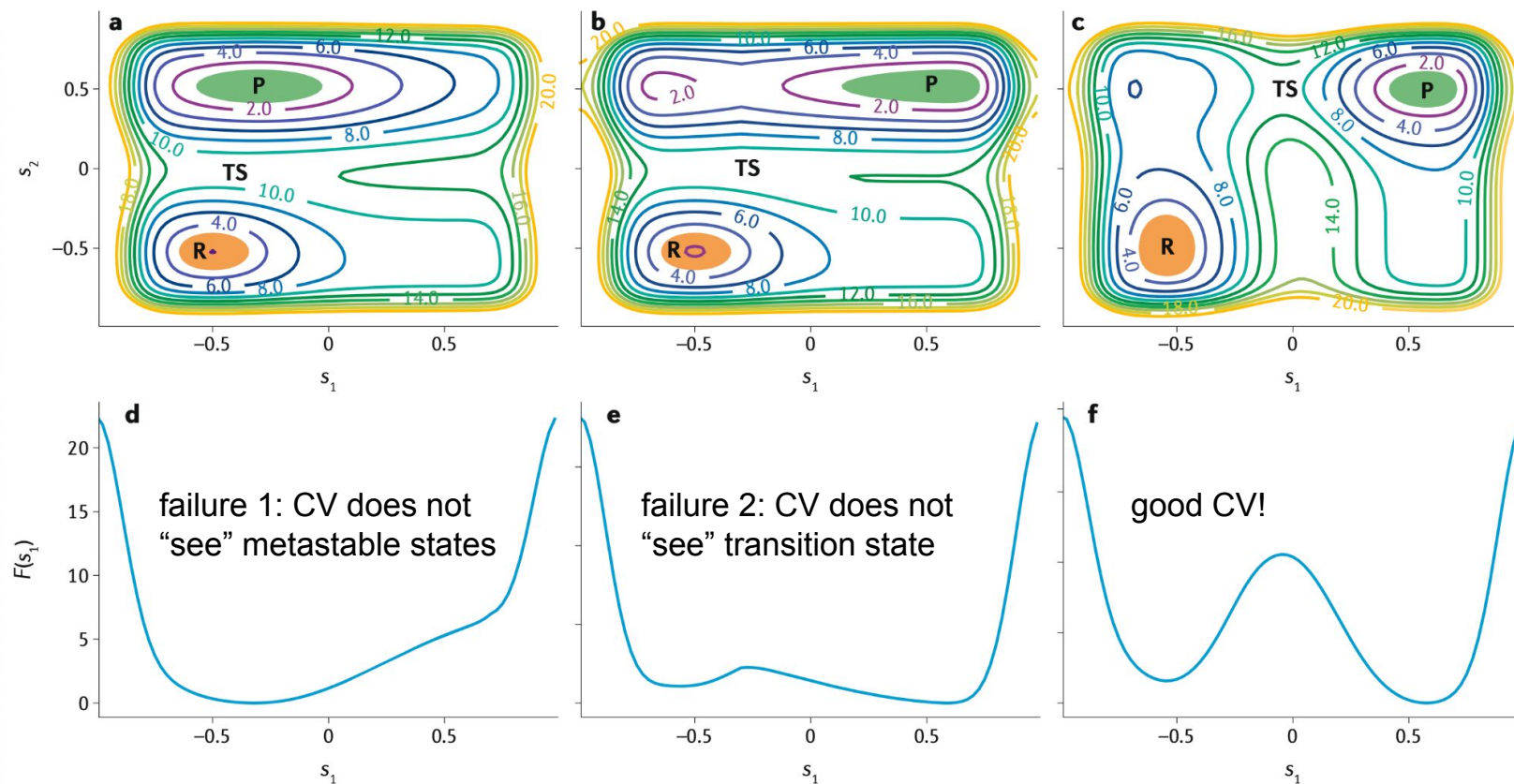


Actually the Gaussian “ansatz” can be seen as a simple smoothing defined over a single point, which works well even if the CV has more than 1 component

Limitation:

- It relies on the knowledge of a “good CV” (as AUS):
 - It should distinguish the two metastable states
 - It should identify the **transition state among them** as well!

Failure modes – “bad CVs”



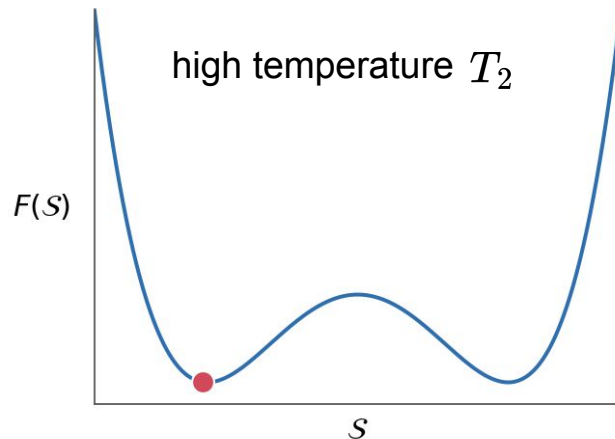
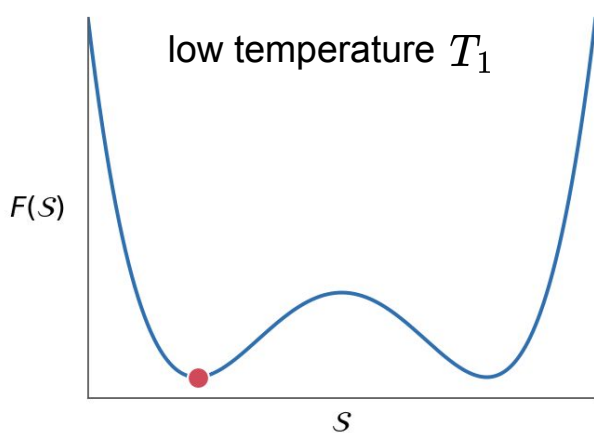
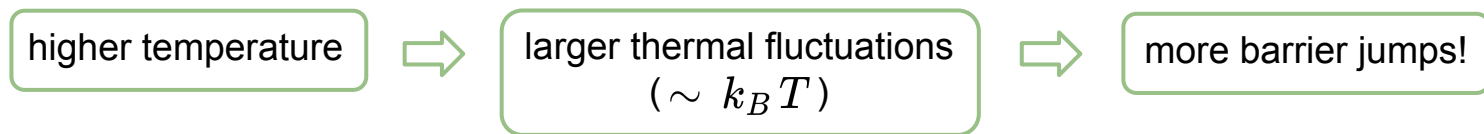
Building a good CV

- Constructing good CVs without prior knowledge on the system is a **challenging problem**
- Some strategies*:
 - Geometry only: Dimensionality reduction (e.g. multidimensional scaling), diffusion maps
 - Kinetics: Time-lagged Independent Component Analysis (TICA), time-lagged autoencoders

*Bussi, G., Laio, A. Using metadynamics to explore complex free-energy landscapes. *Nat Rev Phys* **2** (2020).

Replica-Exchange

- Other method for enhanced sampling; **not CV-based** but base on Monte Carlo Markov chain theory
- Observation: instead of modifying the potential, another “perturbation to the system” that we can perform to enhance transitions is **raising the temperature!**



Replica-Exchange

- Can we sample **only the high-temperature system** and obtain information on the low-temperature one?



in principle yes, with the same reweighting ideas that we use for Umbrella Sampling but a different weight:

$$\langle O \rangle_{T_1} = \frac{\langle O \cdot w \rangle_{T_2}}{\langle w \rangle_{T_1}} \quad w = \exp(-(\beta_2 - \beta_1)U(\mathbf{q}))$$

- Problem behind this idea: most of the high-temperature samples will be assigned **weight equal to zero!** Why?



Increasing the temperature the system starts to explore high-energy configurations which

Replica-Exchange

- Can we use the high-temperature simulation to **improve the ergodicity** of the low-temperature simulation?



this is the idea behind **Replica-Exchange**:

- simulate at different temperatures in a “parallel” manner
 - try to “swap” the states of the two simulations
-
- The “swaps” are based on the theory of Markov chain Monte Carlo

Markov-chain Monte Carlo

- Suppose we want to sample a target distribution $\mathcal{P}_{\text{target}}(\Gamma)$
- Let's recall the definition of a Markov-chain sampler for $\mathcal{P}_{\text{target}}(\Gamma)$ (from day 1):

A Markov-chain sampler of a distribution $\mathcal{P}_{\text{target}}(\Gamma)$ is an update rule $\Gamma \mapsto \Gamma'$ which should satisfy the following properties:

1. **Stationarity of the target distribution:** applied to $\{\Gamma_i\} \sim \mathcal{P}_{\text{target}}$ it returns $\{\Gamma'_i\} \sim \mathcal{P}_{\text{target}}$
2. **Ergodicity:** no “unexplored” regions in the Γ space after a finite number of updates
3. [optional] **Fast decorrelation of samples:** $\mathcal{P}_{\text{target}}(\Gamma)$ sampled with short trajectories!

- Let's model our update rule with a stochastic **transition matrix** $\Pi_{\Gamma \rightarrow \Gamma'}$

Markov-chain Monte Carlo

- Let's model our update rule with a **stochastic transition matrix** $\Pi_{\Gamma \rightarrow \Gamma'}$

$\Pi_{\Gamma \rightarrow \Gamma'}$ is the probability of jumping from Γ to Γ' at each step of the Markov process

$$\Pi_{\Gamma \rightarrow \Gamma'} = P(\Gamma', t + 1 \mid \Gamma, t)$$

Properties of this matrix?

- $\Pi_{\Gamma \rightarrow \Gamma'} \geq 0$ for each Γ, Γ'
- $\sum_{\Gamma'} \Pi_{\Gamma \rightarrow \Gamma'} = 1$ (called **row-stochastic matrix**)

Markov-chain Monte Carlo

- Let's model our update rule with a **stochastic transition matrix** $\Pi_{\Gamma \rightarrow \Gamma'}$

$\Pi_{\Gamma \rightarrow \Gamma'}$ is the probability of jumping from Γ to Γ' at each step of the Markov process

$$\Pi_{\Gamma \rightarrow \Gamma'} = P(\Gamma', t + 1 \mid \Gamma, t)$$

- Such a transition matrix defines the following “master equation”, that tells us what probability we are sampling at time $t+1$ given the probability we were sampling at time t :

$$P_{t+1}(\Gamma) = P_t(\Gamma) + \underbrace{\sum_{\Gamma'} P_t(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma}}_{\text{“gain term”}} - \underbrace{\sum_{\Gamma'} P_t(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}}_{\text{“loss term”}}$$

Markov-chain Monte Carlo

$$P_{t+1}(\Gamma) = P_t(\Gamma) + \sum_{\Gamma'} P_t(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} - \sum_{\Gamma'} P_t(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$$

- Remember: our goal is to use the stochastic update embedded in $\Pi_{\Gamma \rightarrow \Gamma'}$ to sample a specific target equilibrium distribution



We would like $P_{t+1}(\Gamma) = P_t(\Gamma)$ (stationarity)



Let's drop the time index in the master equation:

$$\cancel{P(\Gamma)} = \cancel{P(\Gamma)} + \sum_{\Gamma'} P(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} - \sum_{\Gamma'} P(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$$

$$\sum_{\Gamma'} P(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} = \sum_{\Gamma'} P(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$$

Stationarity condition
(or **balance** condition)

Markov-chain Monte Carlo

$$\sum_{\Gamma'} P(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} = \sum_{\Gamma'} P(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$$

Stationarity condition
(or **balance** condition)

- This is what we need to sample our target distribution: design a stochastic jump “rule” which satisfies this condition

What about **ergodicity**? We also should design $\Pi_{\Gamma \rightarrow \Gamma'}$ such that after a finite number of iterations, there is a nonzero probability of reaching each Γ' from each Γ

- A way to satisfy the stationarity condition above is to impose the equality for each term in the sums:

$$P(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} = P(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$$

Detailed balance condition

Markov-chain Monte Carlo

$$P(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} = P(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$$

Detailed balance condition

- Detailed balance is a **much stronger condition** than balance / stationarity, but it's easier to create a transition matrix with this property!
- Markov-chain Monte Carlo: write $\Pi_{\Gamma \rightarrow \Gamma'}$ as a product of two terms:

$$\Pi_{\Gamma \rightarrow \Gamma'} = M_{\Gamma' \rightarrow \Gamma} \alpha_{\Gamma \rightarrow \Gamma'}$$

“Proposal” matrix
(a stochastic matrix)

“Acceptance” matrix (a simple probability,
between 0 and 1)

- Idea behind this split: first we propose a move (update), then we decide whether to accept it or discard it

Markov-chain Monte Carlo

$$\Pi_{\Gamma \rightarrow \Gamma'} = M_{\Gamma' \rightarrow \Gamma} \alpha_{\Gamma \rightarrow \Gamma'}$$

- Let's substitute this inside the detailed balance condition, $P(\Gamma') \Pi_{\Gamma' \rightarrow \Gamma} = P(\Gamma) \Pi_{\Gamma \rightarrow \Gamma'}$

$$P(\Gamma') M_{\Gamma' \rightarrow \Gamma} \alpha_{\Gamma' \rightarrow \Gamma} = P(\Gamma) M_{\Gamma \rightarrow \Gamma'} \alpha_{\Gamma \rightarrow \Gamma'}$$

$$\frac{\alpha_{\Gamma \rightarrow \Gamma'}}{\alpha_{\Gamma' \rightarrow \Gamma}} = \frac{P(\Gamma') M_{\Gamma' \rightarrow \Gamma}}{P(\Gamma) M_{\Gamma \rightarrow \Gamma'}}$$

it satisfies this:

- Metropolis-Hasting rule:**

$$\alpha_{\Gamma \rightarrow \Gamma'} = \min \left(1, \frac{P(\Gamma') M_{\Gamma' \rightarrow \Gamma}}{P(\Gamma) M_{\Gamma \rightarrow \Gamma'}} \right)$$

Markov-chain Monte Carlo

- **Metropolis-Hasting rule:** $\alpha_{\Gamma \rightarrow \Gamma'} = \min \left(1, \frac{P(\Gamma') M_{\Gamma' \rightarrow \Gamma}}{P(\Gamma) M_{\Gamma \rightarrow \Gamma'}} \right)$

The ratio between the probabilities is extremely useful because it allows to compute this even if we don't know the normalization constants!!

- Can be further simplified: if we take a **symmetric proposal matrix**, $M_{\Gamma' \rightarrow \Gamma} = M_{\Gamma \rightarrow \Gamma'}$

$$\alpha_{\Gamma \rightarrow \Gamma'} = \min \left(1, \frac{P(\Gamma')}{P(\Gamma)} \right)$$

Markov-chain Monte Carlo

Algorithm Markov-chain Monte Carlo (Metropolis–Hastings, symmetric proposal)

Input: target density $P(\Gamma)$, symmetric proposal $T(\Gamma \rightarrow \Gamma') = T(\Gamma' \rightarrow \Gamma)$, initial state Γ , number of steps n_{steps}

```
1: chain  $\leftarrow$  empty_array( $n_{\text{steps}} + 1$ )
2: chain[0]  $\leftarrow \Gamma$  ▷ initial condition: first sample

3: for  $i = 1, \dots, n_{\text{steps}}$  do
4:    $\Gamma' \leftarrow \text{trial}(\Gamma)$  ▷ draw from  $T(\Gamma \rightarrow \cdot)$ 
5:    $\alpha \leftarrow \frac{P(\Gamma')}{P(\Gamma)}$  ▷ symmetric  $T$ : proposal ratio cancels
6:   if rand()  $< \alpha$  then ▷ rand()  $\sim \mathcal{U}[0, 1)$ 
7:      $\Gamma \leftarrow \Gamma'$  ▷ accept
8:   end if
9:   chain[i]  $\leftarrow \Gamma$  ▷ store also on rejection
10: end for

11: return chain
```

Replica-Exchange

- Let's run two NVT simulations with the same parameters, only the temperatures change:

$$T_1 < T_2$$

- The sampled distributions (let's forget the kinetic part) are (e.g. canonical ensemble):

$$\underbrace{\mathcal{P}_1(\mathbf{q}_1) \propto \exp(-\beta_1 \mathcal{U}(\mathbf{q}_1))}_{\text{doesn't explore the second local minimum}} \quad \underbrace{\mathcal{P}_2(\mathbf{q}_2) \propto \exp(-\beta_2 \mathcal{U}(\mathbf{q}_2))}_{\text{explores the second local minimum, but not at the temperature we want!}}$$

- We treat the two replicas as a single “larger” system, with configuration $\mathbf{Q} = (\mathbf{q}_1, \mathbf{q}_2)$
- At each step the two simulations are **independent**: the joint probability we are sampling is just a product of the two,

$$\mathcal{P}(\mathbf{Q}) = \mathcal{P}_1(\mathbf{q}_1) \mathcal{P}_2(\mathbf{q}_2)$$

Replica-Exchange

Idea: enhance the ergodicity of replica 1 by proposing “swap moves” to replica 2, in such a way that the sampled probability $\mathcal{P}(\mathbf{Q}) = \mathcal{P}_1(\mathbf{q}_1) \mathcal{P}_2(\mathbf{q}_2)$ is preserved



Using Markov chain MC step with Metropolis-Hasting rule!

- Symmetric proposal move: every few steps, $\mathbf{Q} = (\mathbf{q}_1, \mathbf{q}_2) \mapsto \mathbf{Q}' = (\mathbf{q}_2, \mathbf{q}_1)$
- Acceptance computed with Metropolis-Hasting:

$$\begin{aligned}\alpha_{\mathbf{Q} \rightarrow \mathbf{Q}'} &= \min \left(1, \frac{\mathcal{P}(\mathbf{Q}')}{\mathcal{P}(\mathbf{Q})} \right) = \min \left(1, \frac{\mathcal{P}_1(\mathbf{q}_2) \mathcal{P}_2(\mathbf{q}_1)}{\mathcal{P}_1(\mathbf{q}_1) \mathcal{P}_2(\mathbf{q}_2)} \right) \\ &= \min \left(1, \exp \left((\beta_2 - \beta_1) (U(\mathbf{q}_2) - U(\mathbf{q}_1)) \right) \right)\end{aligned}$$

Replica-Exchange

Problems of replica-exchange:

- 1) If we only take two replicas at distant temperatures, the acceptance probability tends to be zero!



Reason of this: the distributions of the potential energy become **far apart** with significant temperature differences

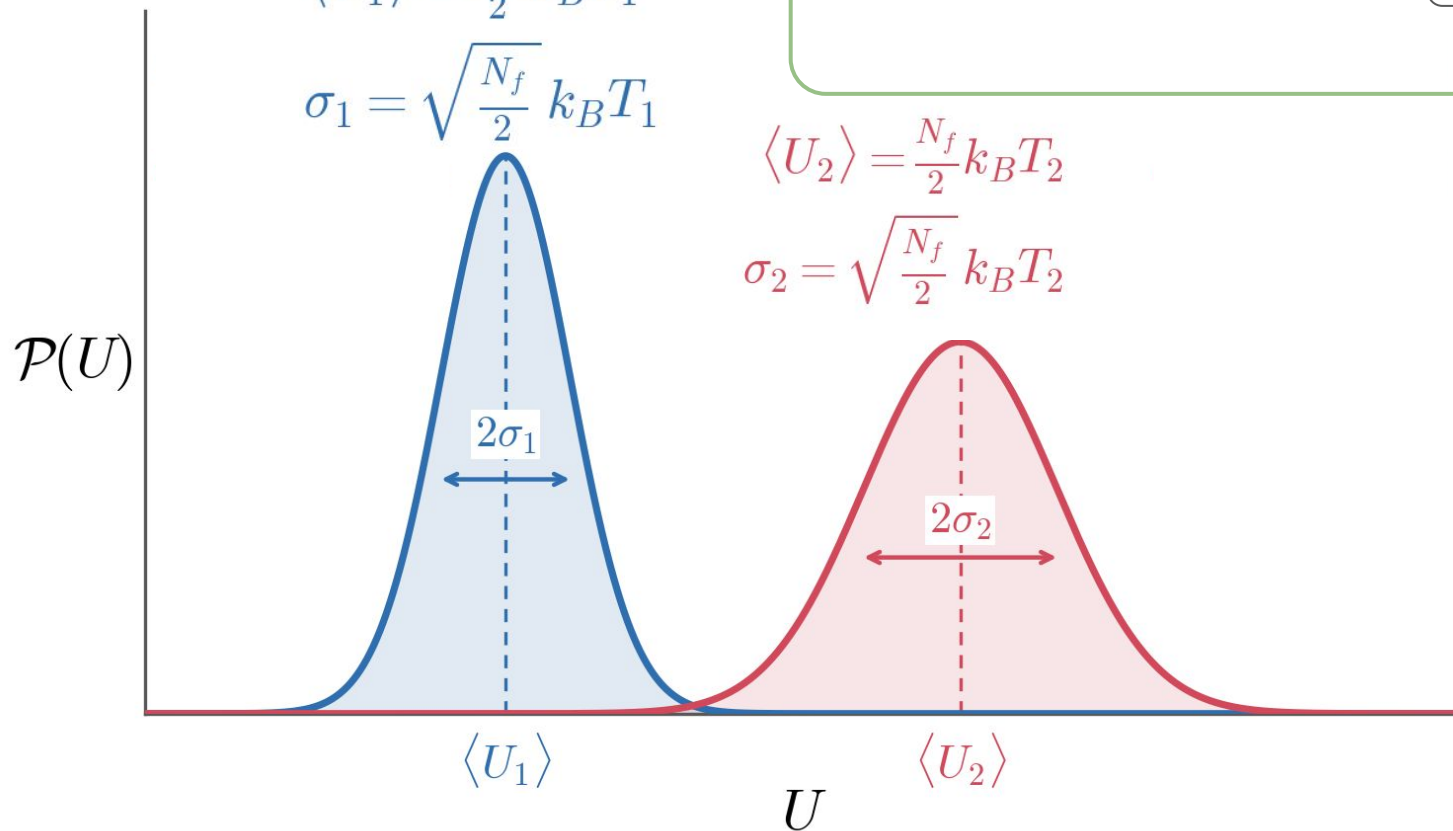
$$\langle U_1 \rangle = \frac{N_f}{2} k_B T_1$$

$$\sigma_1 = \sqrt{\frac{N_f}{2}} k_B T_1$$

$$\alpha_{\text{swap}} = \min \left(1, \exp \left((\beta_2 - \beta_1) \underbrace{(U(\mathbf{q}_2) - U(\mathbf{q}_1))}_{\text{too large!}} \right) \right)$$

$$\langle U_2 \rangle = \frac{N_f}{2} k_B T_2$$

$$\sigma_2 = \sqrt{\frac{N_f}{2}} k_B T_2$$



Replica-Exchange

Problems of replica-exchange:

- 1) If we only take two replicas at distant temperatures, the acceptance probability tends to be zero!

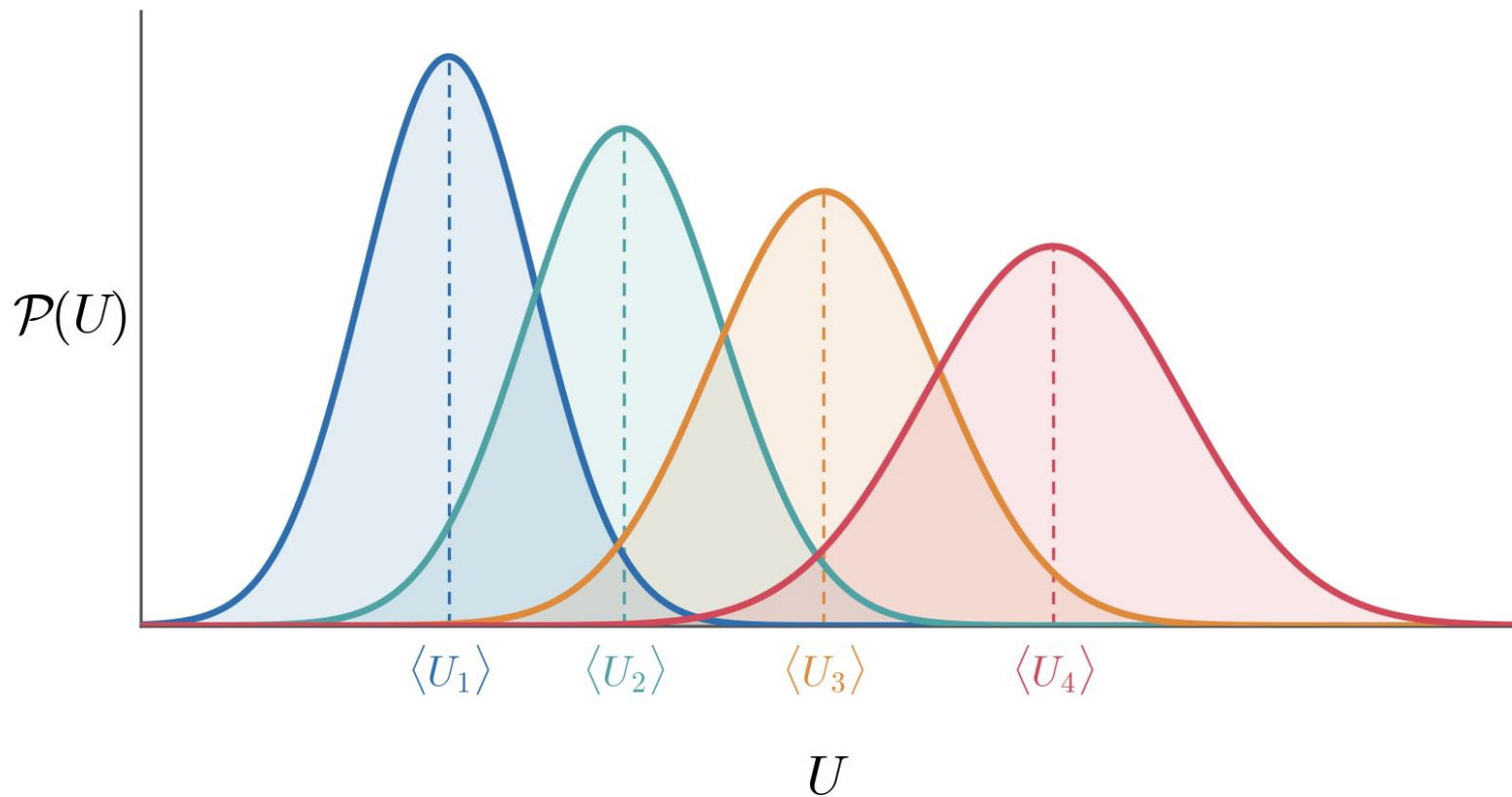


Reason of this: the distributions of the potential energy become **far apart** with significant temperature differences



Solution: running more than two replicas with more similar temperatures, and propose swaps between close-temperature replicas only!

Replica-Exchange



Replica-Exchange

Problems of replica-exchange:

2) Processes sampled at high-temperature can be be very different and more “extreme” to those we observe at low temperature



we sample processes we are not interested in! Reason why this cannot be used for enhancing protein folding